Chapter 12 — Nitrogen and Sulfur

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Nitrogen and ammonia

Q.1 Explain the inertness of nitrogen and the basicity and preparation of ammonia.

Nitrogen is relatively unreactive because the N≡N triple bond is very strong and the molecule is non-polar. The textbook gives the bond enthalpy as +944 kJ mol^{-1} in this chapter. High temperature, lightning or catalysts are needed to make nitrogen react.

Ammonia, NH_3, is a weak base because it accepts a proton to form NH_4^+. The ammonium ion is tetrahedral because nitrogen forms four equivalent bonds and has no lone pair. In the laboratory, ammonia is prepared by heating an ammonium salt with a strong base: $2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$.

Textbook data check: Chapter 6 Table 6.1 gives N≡N as 945 kJ mol^{-1}, while Chapter 12 gives +944 kJ mol^{-1}. Use the value supplied in the question.

Important Conceptual Short Questions — Topic 1

1. Why is nitrogen used as a protective blanket for hydrocarbon shipments?

Nitrogen is unreactive and excludes oxygen, reducing the risk of oxidation or fire.

2. How does ammonia act as a Bronsted-Lowry base?

The lone pair on nitrogen accepts a proton, forming NH_4^+.

3. Why is the ammonium ion tetrahedral?

Nitrogen in NH_4^+ has four bond pairs and no lone pair, so the electron pairs arrange tetrahedrally.

4. How can ammonia be prepared from an ammonium salt?

Heat the ammonium salt with a strong base such as $Ca(OH)_2$ or NaOH. The base removes a proton from NH_4^+, releasing NH_3 gas.

Topic 2 — Oxides of nitrogen, smog and catalytic converters

Q.2 Describe oxides of nitrogen, their sources, photochemical smog, PAN formation and catalytic converters.

Important oxides of nitrogen include N_2O, NO, NO_2, N_2O_4 and N_2O_5. NO and NO_2 are collectively called NOx. They form naturally during lightning and in soils, and by human activities in vehicle engines and power plants where high temperature makes N_2 and O_2 react.

In sunlight, NOx and volatile organic compounds form photochemical smog containing ozone, aldehydes and peroxyacyl nitrates (PANs). NO_2 gives a brown colour. Catalytic converters use Pt, Pd and Rh to convert CO, NOx and unburnt

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hydrocarbons into CO_2, N_2 and H_2O.

Important Conceptual Short Questions — Topic 2

1. Why does NO_2 give a brown colour to photochemical smog?

NO_2 is a reddish-brown gas, so its presence colours the smog brown.

2. How are PANs formed from hydrocarbons and NOx?

Ozone and radicals oxidize hydrocarbons to acyl radicals, which react with oxygen and then NO_2 to form peroxyacyl nitrates.

3. Why are catalytic converters fitted in vehicles?

They catalyse redox reactions that change poisonous CO, NOx and unburnt hydrocarbons into less harmful gases before they leave the exhaust.

4. How does high-temperature combustion produce NO?

At high temperature, nitrogen and oxygen in air gain enough energy to react: $N_2 + O_2 \rightarrow 2NO$.

Topic 3 — Nitrification, denitrification and sulfur

Q.3 Explain nitrification and denitrification, and describe the reactivity and oxidation states of sulfur.

Nitrification is the bacterial oxidation of ammonia or ammonium to nitrite and then nitrate. Denitrification is the bacterial reduction of nitrate to nitrogen gas. Nitrification increases available nitrate for plants, while denitrification returns nitrogen to the atmosphere.

Sulfur exists as S_8 rings and is fairly unreactive at room temperature. It shows oxidation states –2, +4 and +6, as in H_2S, SO_2 and SO_3/H_2SO_4. It burns in oxygen to form SO_2 and reacts with metals to form sulfides.

Important Conceptual Short Questions — Topic 3

1. How does nitrification increase soil fertility?

It converts ammonium into nitrate, a form of nitrogen that plants can absorb readily.

2. Why does denitrification reduce soil nitrate?

Denitrifying bacteria convert nitrate into nitrogen gas, which escapes to the atmosphere and is no longer available to plants.

3. Why does sulfur show variable oxidation states?

Sulfur can gain two electrons to complete its octet or share/lose electrons with more electronegative elements such as oxygen, giving –2, +4 and +6 states.

4. How does sulfur behave as an oxidizing agent with metals?

Sulfur accepts electrons from metals to form sulfide ions, S^{2-}, so the metal is oxidized and sulfur is reduced.

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Topic 4 — Uses of sulfur and sulfuric acid

Q.4 Describe the uses of sulfur and the manufacture and properties of sulfuric acid.

Sulfur is used in vulcanization of rubber, fertilizers, gunpowder, fungicides and organic synthesis. Vulcanization forms sulfur cross-links between rubber chains, improving elasticity, strength and resistance to heat.

Sulfuric acid is manufactured by the Contact process: sulfur or sulfide ore is burned to SO_2, SO_2 is oxidized to SO_3 over V_2O_5, and SO_3 is absorbed in concentrated H_2SO_4 to form oleum, $H_2S_2O_7$, which is diluted carefully. H_2SO_4 is a strong acid, dehydrating agent and oxidizing agent when hot and concentrated. It is used in fertilizers, batteries, dyes, detergents and explosives.

Textbook check (p. 277): iron pyrite is printed as FeS in the Contact-process description; the common formula of iron pyrite is FeS_2. Use the formula intended by the question and balance accordingly.

Important Conceptual Short Questions — Topic 4

1. Why is SO_3 absorbed in concentrated H_2SO_4 instead of water?

Direct reaction with water is highly exothermic and produces a fine acid mist. Absorption in concentrated acid forms oleum, which can be diluted safely.

2. How does vulcanization improve rubber?

Sulfur cross-links connect polymer chains, making rubber stronger, more elastic and less sticky over a wider temperature range.

3. Why is sulfuric acid called a dehydrating agent?

It has a strong affinity for water and can remove hydrogen and oxygen from compounds in the ratio of water, as in charring sugar.

4. How does the Contact process use a compromise temperature?

Low temperature favours SO_3 yield but slows the rate. About 400–450 °C with V_2O_5 gives a useful rate and good yield.

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